Chapter 7: Sampling Distributions

7.1: Introduction

7.2: Distributions Related to the Normal Distribution

7.3: Central Limit Theorem

7.1: Introduction

Recall:

Statistical Inference: Drawing conclusions about ***populations*** (**parameters**) based on information collected in ***samples*** (**statistics**).

Statistical inference goes as follows:

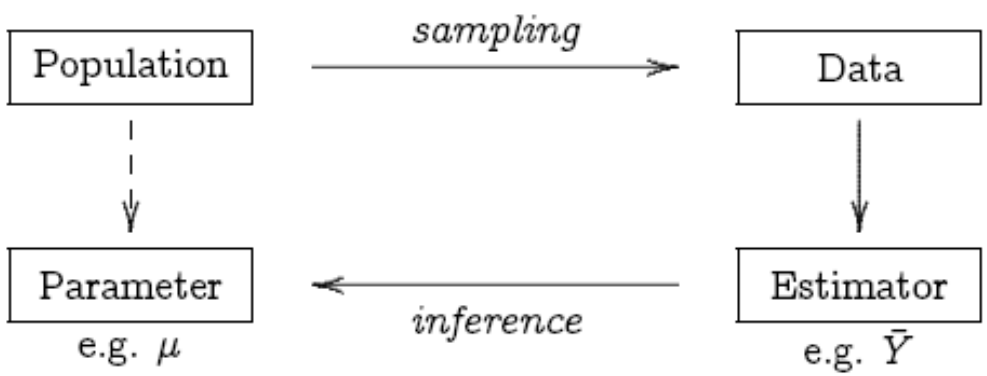


Illustration :

Suppose our goal is to answer a question about some population mean μ.

To get started we can collect some information in the form of a sample and estimate the paramater. Let's consider the sample mean as our estimate.

If $Y_1, Y_2, ..., Y_n$ represents our random sample from this population, where $Y_1, Y_2, ..., Y_n$ are identically distributed with $E[Y_i] = \mu,\ i = 1, ..., n,$ the sample mean is $\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (a function of the random variables in the sample).

Parameter : μ - the (unknown) population mean

Estimator : $\overline{Y}$ - the sample mean

The key to inference is understanding the properties of the estimator $\bar{Y}$.

Questions to consider:

What does the sample imply about the population? Or, what does the estimate imply about the paramater.

- Is $\bar{Y}$ equal to μ?
- Is $\bar{Y}$ close to μ?
- How close can we expect $\bar{Y}$ to be to μ? With what certainty?

Beyond the questions above, we would also like to know why we are using $\bar{Y}$ to estimate μ? Is that a good choice? The best choice? Are there other estimates we should consider?

These questions will be answered based on the **sampling distribution** of $\bar{Y}$.

In general:

Let $Y_1, Y_2, \ldots, Y_n$, represent a random sample of independent and identically distributed random variables from distribution Y.

Definition 7.1 : Statistic

$\Rightarrow$ A **statistic** is any function of random variables collected in a sample, and known constants. Statistics are random variables.

e.g. $Y_1,\ \overline{Y},\ S^2,\ Y_{(k)},\ Y_{(n)} - Y_{(1)},\ 3Y_1 + e^{Y_2} - \cos(Y_3)$ are all statistics.

Note: We will focus on statistics that will help us estimate our parameters. We wish to describe the **probability distribution of these statistics.**

Definition 7.1+ : Sampling Distribution

$\Rightarrow$ The probability distribution of a **statistic** is called a sampling distribution.

e.g. What is the probability distribution of $\bar{Y}$?

7.2: Distributions Related to the *Normal* Distribution

This section will explore some special statistics based on sampling from a *Normal* population.

Let $Y_1, Y_2, \ldots, Y_n$, represent a random sample of independent and identically distributed $Normal(\mu_Y, \sigma_Y^2)$ random variables.

- $f_Y(y) = \dfrac{1}{\sigma_Y \sqrt{2\pi}} e^{\frac{-1}{2\sigma_Y^2}(y-\mu_Y)^2}, -\infty < y < \infty$
- $m_Y(t) = e^{\mu_Y t + \frac{1}{2}\sigma_Y^2 t^2}$

Theorem 7.1 :

Consider a random sample, $Y_1, \ldots, Y_n$, where $Y_i \sim N(\mu_Y, \sigma_Y^2)$.

Then, the sampling distribution of the sample mean is

$$\Rightarrow \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i \sim Normal(\mu_{\bar{Y}} = \mu_Y,\ \sigma_{\bar{Y}}^2 = \sigma_Y^2 / n).$$

Result: $Z = \dfrac{\bar{Y} - \mu_{\bar{Y}}}{\sigma_{\bar{Y}}} = \dfrac{\bar{Y} - \mu_Y}{\sigma_Y / \sqrt{n}} \sim Normal(0,1).$

Theorem 7.1: Proof

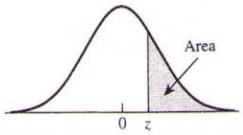
Example 1:

The amount filled (in ounces) by a certain bottling machine is a random variable following approximately a Normal distribution with unknown mean and standard deviation equal to 2 ounce.

a) If a random sample of size n = 9 is gathered, what is the probability that the sample mean will be within 0.3 ounces of the true mean?

b) What is the minimum sample size, n, so that the sample mean will be within 0.3 ounces of the population mean, 90% of the time (upon repeated sampling)?

Table 4 Normal Curve Areas
Standard normal probability in right-hand tail
(for negative values of z, areas are found by symmetry)



	Second decimal place of z									
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0352	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.00135									
3.5	.000 233									
4.0	.000 031 7									
4.5	.000 003 40									
5.0	.000 000 287									

From R. E. Walpole, *Introduction to Statistics* (New York: Macmillan, 1968).

Theorem 7.2 :

Consider a random sample, $Y_1, \ldots, Y_n$, where $Y_i \sim N(\mu_Y, \sigma_Y^2)$.

Then,

$$\Rightarrow \sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n} \left(\frac{Y_i - \mu_Y}{\sigma_Y} \right)^2 \sim \chi_n.$$

Theorem 7.2: Proof

Example 7:

A machine in a heavy-equipment factory produces steel rods of length ***Y***, where ***Y*** is approximately normally distributed with mean 6 feet and variance 0.2. The cost, ***C***, of repairing a rod that is not exactly 6 feet in length is proportional to the square of the error and is given, in dollars, by $C = 4(\boldsymbol{Y} - 6)^2$. If 50 rods with independent lengths are produces in a given day, find

- The expected value and variance for the total cost for repairs.
- The probability that the total cost for repairs in that given day will exceed $48.

Table: Chi-Square Probabilities

The areas given across the top are the areas to the right of the critical value. To look up an area on the left, subtract it from one, and then look it up (ie: 0.05 on the left is 0.95 on the right)

df	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	---	---	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

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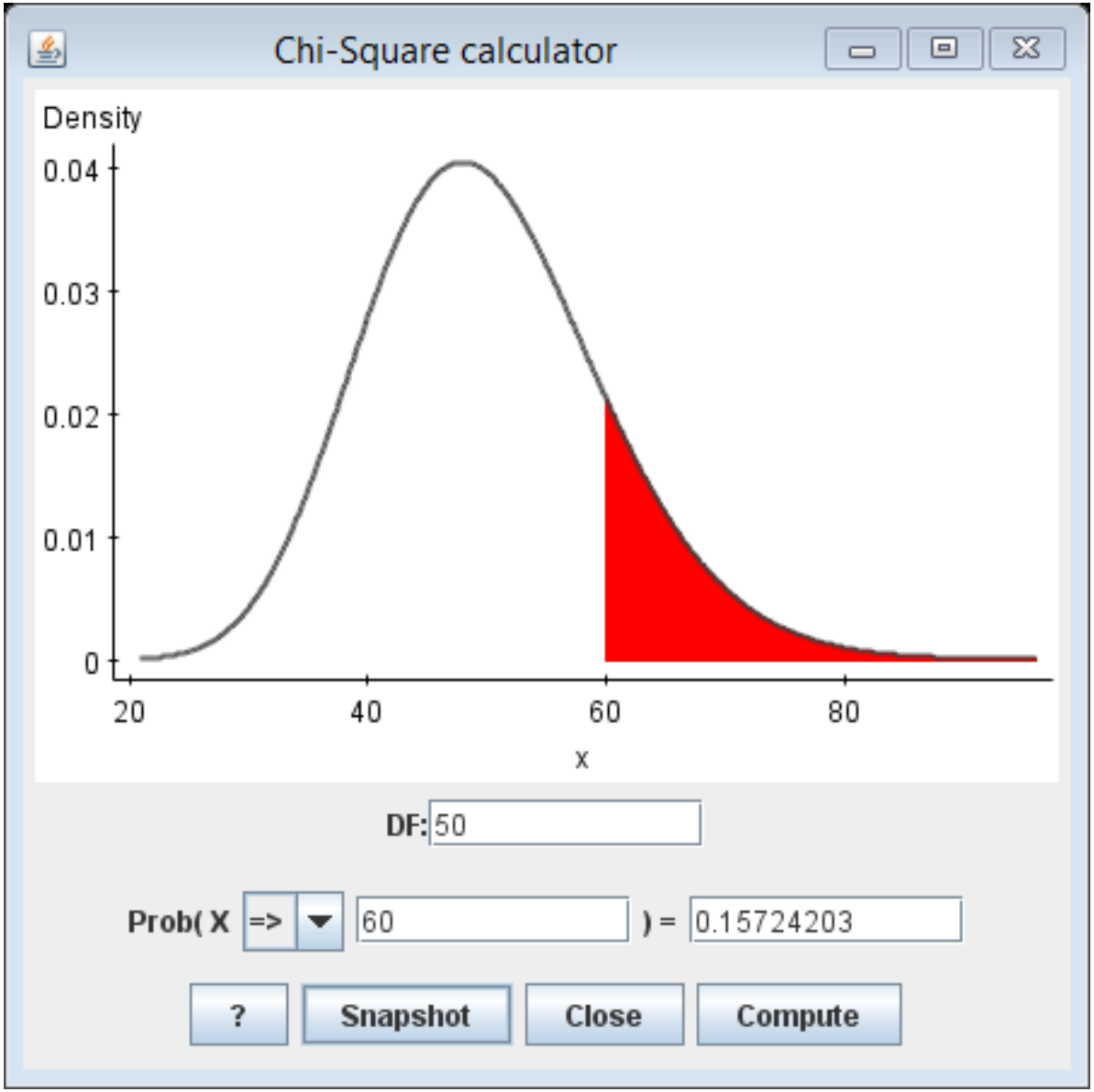

Chi-Square calculator
Density
0.04
0.03
0.02
0.01
0
20
40
60
80
x
DF: 50
Prob(X => 60) = 0.15724203
?
Snapshot
Close
Compute

Example:

Suppose that $Z_1, \ldots, Z_4$ is a random sample from a standard normal distribution. What is $V[U]$, where $U = (Z_1 + 2Z_2 + 2Z_3 + 4Z_4)^2$?

Theorem 7.3 :

Consider a random sample, $Y_1, \ldots, Y_n$, where $Y_i \sim N(\mu_Y, \sigma_Y^2)$.

The sample variance is defined as

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2.$$

Then,

i) $\bar{Y}$ and S^2 are independent random variables.

ii) $$\frac{(n-1)S^2}{\sigma_Y^2} = \frac{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}{\sigma_Y^2} \sim \chi^2_{n-1}.$$

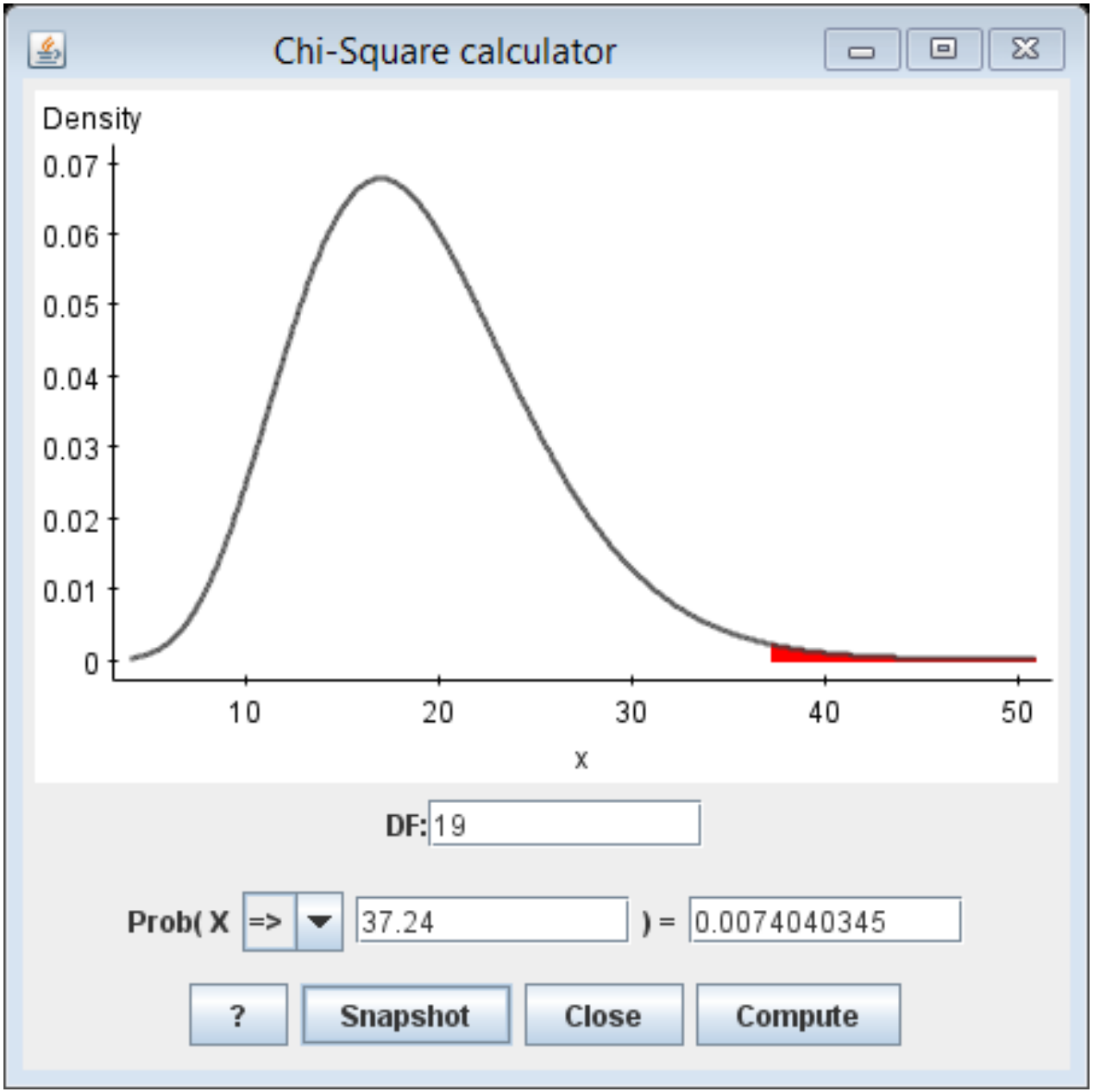
Theorem 7.3ii: Proof

Example 14:

Suppose a process in under control if and only if $\sigma \leq 0.6$. From a random sample of size 20 it was observed that ***s*** = 0.84. Assuming that the observations are *iid* normal random variables; does it appear that the process is out of control?

To help answer this question, find the probability that a sample of size 20 would give a result of ***S*** $\geq$ 0.84, if in fact the process was under control at $\sigma = 0.6$.

(This is an example of a hypothesis test for the population variance.)


Chi-Square calculator
Density
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
10
20
30
40
50
x
DF: 19
Prob(X => 37.24) = 0.0074040345
?
Snapshot
Close
Compute

Definition 7.2 : t Distribution

Consider independent random variables Z and W where

a) $Z \sim N(0,1)$,

b) $W \sim \chi^2_v$.

$\Rightarrow T = \dfrac{Z}{\sqrt{W/v}} \sim t_v$ (A t distribution with v degrees of freedom.)

Properties of t_v :

1. $E[t_v] = 0$

2. $V[t_v] = \dfrac{v}{v-2}, \quad v > 2.$

Result 7.2 : t Distribution

Consider a random sample, $Y_1, \ldots, Y_n$, where $Y_i \sim N(\mu_Y, \sigma_Y^2)$.

Then,

$$\Rightarrow T = \frac{\bar{Y} - \mu_Y}{S_Y \big/ \sqrt{n}} \sim t_{n-1}.$$

Proof :

Example 15:

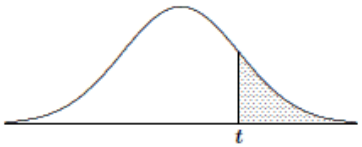
In ***n***=16 one-hour test runs, the gasoline consumption of an engine showed an average of 16.4 gallons with standard deviation 2.1. Evaluate the claim that the true average gasoline consumption of the engine is ***k*** gallons vs. the claim that consumption is over ***k*** gallons, for ***k*** = 12, 14, 16.

To help evaluate these claims, find the probability of observing a sample mean of 16.4 or higher (further from ***k***) if in fact the consumption was equal to ***k***.

(This is an example of a hypothesis test for the population mean.)

Assume gasoline consumption is approximately Normal.

t table



Cutoff probabilities

Degrees of freedom	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.663	.858	1.061	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	.855	1.057	1.314	1.703	2.052	2.150	2.473	2.771	3.057	3.421	3.690
28	0.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	.849	1.047	1.295	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	0.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291

Example:

Suppose that $Z_1, \ldots, Z_n$ are independent standard normal random variables.

If $V = \dfrac{c(Z_1 + Z_2)}{\left(Z_3^2 + Z_4^2 + Z_5^2\right)^{1/2}} \sim t_{df}$, what are c and df ?

Definition 7.3 : F Distribution

Consider independent random variables W_1 and W_2 where

a) $W_1 \sim \chi^2_{v_1}$,

b) $W_2 \sim \chi^2_{v_2}$.

$\Rightarrow F = \dfrac{W_1 / v_1}{W_2 / v_2} \sim F^{v_1}_{v_2}$ (An F distribution with v_1 and v_2 degrees of freedom.)

Properties of $F^{v_1}_{v_2}$:

1. $E[F^{v_1}_{v_2}] = \dfrac{v_2}{v_2 - 2}$, $v_2 > 2$.

2. $V[F^{v_1}_{v_2}] = \dfrac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)}$, $v_2 > 4$.

Result 7.3 : F Distribution

Consider a random sample of size n_1 from population Y_1, where $Y_1 \sim N(\mu_1, \sigma_1^2)$.

Consider a random sample of size n_2 from population Y_2, where $Y_2 \sim N(\mu_2, \sigma_2^2)$.

Then,

$$\Rightarrow F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \sim F_{n_2-1}^{n_1-1}.$$

Example:

Suppose that $Z_1, \ldots, Z_n$ are independent standard normal random variables.

If $V = c\dfrac{(Z_1 - Z_2)^2}{Z_3^2 + Z_4^2 + Z_5^2 + Z_6^2} \sim F_{v_2}^{v_1}$, what are c, v_1, and v_2?